Gravitational Time Dilation as an Information Load–Capacity Saturation Curve: Exact Rewritings and a Computational Reading

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Abstract

We show that the time-dilation factor of the classical black-hole family of general relativity admits an exact rewriting as a saturation curve of an information-theoretic load ratio: $\lambda^2 = 1 - S_{load}/S_{max}$, where S_{load} is the Bekenstein bound of the enclosed energy and S_{max} the holographic capacity of the enclosing surface; the horizon is exactly the point of full saturation ($\lambda \to 0$ as load → capacity). The rewriting is unified across the family. For Kerr, the load ratio factorizes exactly as $x = x_{pole}\cdot\gamma^2$: the θ-dependence of time dilation is carried entirely by the Lorentz factor γ of the frame-rotation velocity, and the horizon saturates uniformly at every angle. For Kerr–Newman the identity holds exactly, with charge entering through the enclosed-energy accounting fixed by classical electrodynamics. For static RN–dS it holds exactly, with the de Sitter horizon as a capacity-exhaustion surface (the KN–dS angular sector carries an $O(\Lambda a^2)$ gap, located and reported). The load definition itself differs by case — strict Bekenstein for Schwarzschild, Brown–York-type quasi-local for KN, horizon-anchored algebraic decomposition for Kerr — while the rewriting form is the same; this trichotomy (§2) delimits the precise content of the "Bekenstein anchoring".

Two further layers are offered. Structurally (Layer II), the curve has the *form* of the throttling response of a resource-limited computing system running at its stability margin. Interpretively (Layer III), a three-rule computational engine — marginality, network causality, drag conservation — reproduces the curve to machine precision for Kerr and Kerr–Newman; it is a structured reconstruction with enumerated inputs (§6.7), not an independent derivation, and the KN match is a constrained consistency check, not an independent prediction.

Boundaries. The rewriting is, mathematically, a restatement with no new observational content. The no-drop and full-budget policies behind Rule A are model choices, not logical necessities; no feasible experiment distinguishes a simulated universe from a substrate one; the strong ("omnipotent simulator") version of the hypothesis is unfalsifiable by construction. Within these bounds, the framework converts the simulation hypothesis's central intuition from analogy into a quantitative hypothesis whose toy-model-specific predictions are testable in analogue systems — a computational reading of the GR curve that the substrate reading remains free to decline.

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1 Introduction

1.1 The resource argument

The simulation hypothesis in its scientific form rests on a single economic premise: *computation is expensive*. A simulator of a universe must therefore budget, and the natural budget policy of any rendering system is familiar: simulate in full detail only what is observed, degrade what is not, and, when a region's demand approaches the budget, **slow the region's clock rather than drop frames** — dropped frames are visible from the inside as violations of the very laws the simulator is paid to maintain; a smoothly slowed clock is invisible.

If the premise holds, then inside the simulation one should observe a monotone relation between a region's information content and its local clock rate, with the clock freezing exactly where the information content saturates the region's capacity. The purpose of this paper is to point out that general relativity exhibits precisely this relation — in an exact, parameter-free form — and to characterize how far a purely computational ruleset can account for it.

1.2 Prior work

The idea that time dilation reflects processing load has a literature, but at the qualitative level: Whitworth [21] argued that "interactions are processing overloads" in a virtual-reality picture of physics; the HTUM framework [30] writes a curve $\tau = 1/\sqrt{(1 - 2C/C_max)}$ with a model-specific "causal processing function" C (non-peer-reviewed); Verlinde [18] derived Newtonian gravity from entropic screens without touching the time-dilation curve; Lloyd [13, 14] computed the universe's total computational budget without connecting it to local clocks; Beane–Davoudi–Savage [34] searched for lattice artifacts in cosmic rays (an orthogonal route); Bostrom [19] supplied the statistical prior. On the thermodynamic side, the closest programs are Jacobson's derivation of the Einstein equation from horizon thermodynamics [43], Padmanabhan's emergent-gravity program [44], the thermal-time reading of Rovelli and collaborators (temperature as the "speed of time") [56, 57], the gravitational-time-dilation decoherence of Pikovski et al. [58], the GR–Landauer connection [50, 51], and the deterministic/cellular-automaton reading of quantum mechanics of 't Hooft [46] — all of which treat information as primary but none of which writes the time-dilation factor itself as a load–capacity curve. The ZAMO kinematics used in §5 are standard [52], as are the gravitomagnetic clock-effect results collected in Theorem 4 [53, 54]; what is new there is the load-ratio interpretation. No prior work, to our knowledge, has anchored the load ratio in the Bekenstein bound and holographic capacity, proved the exact correspondence for Schwarzschild and Kerr, or built a structured reconstruction with enumerated inputs that reproduces the curve. We note in passing that a growing non-peer-reviewed literature (self-published and preprint venues) shares the general "saturation/latency" framing, in a few cases with formally similar curves; none of these works provides the anchored identity, the Kerr decomposition, or a predictive test. A detailed prior-work differentiation table is available from the authors. Separately, bifurcation-theoretic

normal forms have recently entered GR through the program of Cotsakis and collaborators — saddle-node structure of cosmic acceleration, bifurcation diagrams for singularities and black holes [47–49] — and our Rule A (§6.1) applies the same mathematics to a different object, the local clock rate, with the relation flagged there.

1.3 Reading guide: the three layers

Every claim in this paper belongs to exactly one of three layers, and we ask the reader to keep them separate:

- **Layer I (mathematical)** — the identities of §§3–4 are *restatements* of standard geometry. "The time dilation factor equals a load–capacity saturation curve" is trivially true in this layer and adds nothing to the observational content of GR.
- **Layer II (structural)** — the curve has the *form* of the throttling response of a resource-limited system at its stability margin, with saturation exactly at the maximum information density. This is a non-trivial observation about the structure of GR's solutions, independent of any simulator.
- **Layer III (interpretive)** — the claim that this coincidence is a signature of resource-constrained computation. Only this layer is conjectural; it is the layer tested by the generative ruleset of §6–7 and disciplined by the falsification conditions of §9.

1.4 Summary of results

(i) Schwarzschild: $\lambda^2 = 1 - S_{load}/S_{max}$ exactly (Theorem 1); (ii) the relevant load is *cumulative* — the Newtonian potential, not the local density (Theorem 2, GPS as discriminator); (iii) Kerr: the identity survives rotation for zero-angular-momentum observers, with the load ratio factorizing into a polar part times γ^2 of the frame-rotation velocity; the horizon saturates uniformly at all angles (Theorems 3–4); (iv) a three-rule computational engine with enumerated inputs reproduces all of the above to machine precision, and passes one constrained consistency check (KN, §7.1); (v) consistency: the horizon's erasure is Landauer-saturated with zero excess (§8.1); (vi) epistemology: the weak version and its falsification conditions (§9).

In one sentence: general relativity's time-dilation curve rewrites exactly as a load–capacity saturation curve — Kerr's θ-dependence lives entirely in the gear-factor Lorentz term, the horizon saturates uniformly, and the load definition differs across Schwarzschild/KN/Kerr while the rewriting form is unified.

2 Notation and conventions

Geometric units $G = c = 1$ unless stated; entropies in nats; $\ell_P^2 = G\hbar/c^3$. For a spherical region of radius R enclosing energy E:

- **Load** $S_{load}(R) := 2\pi RE/\hbar c$ — the Bekenstein bound [7] of the enclosed energy;

- **Capacity** $S_{max}(R) := A/4\ell_P^2 = \pi R^2/\ell_P^2$ — the holographic capacity [9, 10, 15] of the bounding surface;
- **Load ratio** $x := S_{load}/S_{max}$; **clock rate** $\lambda := d\tau/dt$.

Kerr–Newman-(A)dS (Boyer–Lindquist): $\Sigma = r^2 + a^2\cos^2\theta$; $\Delta = (r^2+a^2)(1 - \Lambda r^2/3) - 2Mr + Q^2$, with Λ the physical cosmological constant; $\rho^2 = \Sigma(r^2+a^2) + a^2\sin^2\theta(2Mr - Q^2)$ (equivalently $(r^2+a^2)^2 - a^2\Delta\sin^2\theta$ for $\Lambda = 0$). Throughout, E is the ADM/Christodoulou mass parameter M, and the "enclosed energy" used in §7.1 is the quasi-local (Brown–York-type) energy — the ADM mass minus the electromagnetic energy stored outside the sphere — which for KN gives $E(r) = M - Q^2/(2r)$. The status of the load accounting differs by case, and we state it precisely: for Schwarzschild, the load is the strict Bekenstein bound of the enclosed energy; for KN, it is the Brown–York-type quasi-local accounting (conditional); for Kerr there is no spherically symmetric enclosed energy, and the load 2Mr per unit solid angle is a **horizon-anchored algebraic decomposition** of the identity $\rho^2 - \Sigma\Delta = 2Mr(r^2+a^2)$ — not a Bekenstein bound of a uniquely defined enclosed energy. This trichotomy is the precise content of the "Bekenstein anchoring" used below.

3 Theorem 1 — the Schwarzschild identity (Layer I)

Theorem 1. For a static spherical source of mass M, the clock rate of a stationary observer at radius R satisfies

$$\lambda^2 = 1 - x, \quad x = S_{load}/S_{max} = r_s/R,$$

where $r_s = 2GM/c^2$. In particular $\lambda \to 0$ at $R = r_s$, where $S_{load} = S_{max} = S_{BH} = 4\pi GM^2/\hbar c$.

Proof. $x = (2\pi RE/\hbar c)(\ell_P^2/\pi R^2) = 2GE/c^4R = r_s/R$; and $\lambda^2 = 1 - r_s/R$ in Schwarzschild geometry. ∎

Remark 1 (billing by the bound). The identity uses the Bekenstein *bound*, not the actual entropy of the enclosed matter (which is typically 10–20 orders of magnitude smaller). The computational reading of §6 supplies the rationale: a simulator that billed by actual entropy would have to know the microstate, and knowing it *is* the simulation; the bound is the only macroscopically measurable load. Section 6.5 turns this argument into a quantitative experiment: billing by a fluctuating actual load produces clock jitter 700× larger than billing by the bound, and the observed smoothness of physical clocks selects the bound.

Remark 2 (area budget). Reproducing the GR curve requires capacity ∝ area; a volume-proportional capacity yields $\lambda^2 = 1 - S/V$, which GR excludes. We state this as a *rephrasing* of the known holographic area law [15, 59], not as an independent derivation of it: because the area law holds, the identity takes area form.

4 Theorem 2 — the load is cumulative (Layer II)

Theorem 2. In the static weak-field limit,

$x = 1 - \lambda^2 \approx -2\varphi/c^2 = (2G/c^2) \int \rho(r')/|r-r'|\, d^3r'$.

The load ratio is the density integrated against $1/|r-r'|$ — the Newtonian potential — not the local density. **Discriminator:** a clock at the Earth's surface sits in matter of density $\sim 10^3$ kg/m³ and ticks *slower* than a GPS satellite surrounded by $\sim 10^{-20}$ kg/m³ — the naive "local density" reading has the wrong sign, the cumulative reading the right sign and magnitude. In the simulator language: a region's frame rate depends on the total scene weight within its causal window, not on the object count under the observer's feet; "render distance = causal horizon" (§5 of the companion plan).

5 Theorems 3–4 — Kerr: rotation is not extra load

Theorem 3 (ZAMO curve). For zero-angular-momentum observers in Kerr [52], with $\rho^2 = \Sigma(r^2+a^2) + 2Ma^2 r \sin^2\theta$,

$\lambda_Z^2 = \Sigma\Delta/\rho^2 = 1 - x$, $x = 2Mr(r^2+a^2)/\rho^2$,

and (i) $x \in [0,1)$ outside the horizon; (ii) **$x \to 1$ as $r \to r_+$ for every θ** — the horizon saturates uniformly; (iii) $a = 0$ recovers $x = 2M/r$. *Numerically verified to 4.4×10^{-16} over 2×10^5 random configurations (§B).*

Theorem 3b (gear factor). The load ratio factorizes exactly:

$x = x_{pole} \cdot \gamma^2$, $x_{pole} = 2Mr/(r^2+a^2)$, $\gamma = (r^2+a^2)/\rho = 1/\sqrt{(1-v^2)}$,

with $v = a\sqrt{\Delta}\sin\theta/(r^2+a^2) = \Omega\sqrt{\Delta}\sin\theta$, $\Omega \equiv a/(r^2+a^2)$. The θ-dependence of Kerr time dilation is *entirely* contained in γ^2; at the horizon Ω reduces to the standard $\Omega_H = a/(2Mr_+) = a/(r_+^2+a^2)$. **The gear factor is a Lorentz factor of the frame-rotation profile.**

Theorem 4 (stationary observers; "rotation is not extra load"). The individual facts (i)–(iv) below are standard — ZAMO/stationary kinematics [52] and the gravitomagnetic clock effect [53, 54] — what is not standard is their unified load-ratio reading. For stationary observers, $\lambda_{stat}^2 = 1 - 2Mr/\Sigma$, whence: (i) on the equator $\lambda_{stat}^2 = 1 - 2M/r$ — identical to Schwarzschild at every radius (where stationary observers exist, i.e., outside the ergosphere; on the equator this means $r > 2M$) and every order in a; (ii) at the poles the clock runs *faster* ($\lambda^2 = 1 - 2Mr/(r^2+a^2)$); (iii) the first spin correction is $O(a^2)$ and positive: $\lambda^2 = 1 - 2M/r + 2Ma^2\cos^2\theta/r^3 + O(a^4)$; the O(a) effect is entirely off-diagonal (frame dragging); (iv) the horizon entropy *decreases* with spin, $S_{BH} = 2\pi Mr_+$, $S_{BH}(a=0.9M)/S_{BH}(0) = 72\%$. Rotation therefore does **not** add clock-slowing load: it adds an off-diagonal frame rotation and reduces the total load. This falsifies the naive "spinning regions render slower" expectation.

6 The generative ruleset (Layer III)

6.1 Rule A — marginality (a natural toy model, not a logical necessity)

A region's frame engine is modeled by the saddle-node normal form $dy/dt = (1-x) - y^2$, with state $y \geq 0$ (frame progress). Two *policy choices* are imposed: **no dropped frames** (the engine must remain bounded: $x \leq 1$) and **maximal throughput** (the budget binds: zero slack, $\mu = 1-x$). We emphasize that these are choices of the model, not constraints every simulator must satisfy — a real engine could instead employ error correction, rollback, hidden latency, or safety slack, and a multi-variable engine could cross other bifurcations (cusp, Bogdanov–Takens). The claim is conditional: **a scalar engine that chooses these two policies** runs at its stable fixed point, and we identify two distinct marginal quantities:

- **clock rate** $\lambda = y = \sqrt{(1-x)}$ *— the sustainable activity level, matching the GR value exactly** (no normalization freedom is invoked);
- **recovery rate** $2\sqrt{(1-x)}$ — the marginal eigenvalue $|f'(y^*)|$, governing the decay of perturbations (§6.6).

Both quantities are $\sqrt{(1-x)}$ up to an O(1) constant; GR's curve corresponds to the clock-rate identification. (Numerically verified to <0.1%; §B.)

Why a saddle-node? We choose the saddle-node as the simplest scalar codimension-one normal form with a stable non-zero fixed point $y = \sqrt{\mu}$*; we do not claim it is uniquely forced by the no-drop constraint. For comparison: the transcritical* $dy/dt = \mu y - y^2$ *has* $y = 0$ *as an* unstable* fixed point for $\mu > 0$ (within the constraint region it is not a dead state); it is set aside as a model choice, as is the pitchfork, whose Z_2-symmetric branch structure a progress variable does not possess. Selecting the $\sqrt{}$ branch is a match to the GR curve, and we acknowledge it as such. The same normal form has recently been identified in symmetry-reduced gravitational dynamics — the saddle-node structure of cosmic acceleration and of spacetime-singularity/bifurcation diagrams [47–49] — and observed experimentally at analogue (white-hole) horizons [55], supporting the broader thesis that GR's reduced flows live in the versal-unfolding landscape; our application differs in target (local clock rates rather than global cosmological flows) but shares the mathematics.

Local versus global validity. The normal-form argument guarantees the $\sqrt{}$ form **near the marginal point** ($x \to 1$) — and this is the precise content of the universality claim. The global linearity $\lambda^2 = 1-x$ for all $x \in [0,1]$ is a further, model-level statement — the engine's budget accounting is linear in the single load parameter x — which GR satisfies exactly and which we flag as a property of the model rather than a theorem. Both readings are kept separate throughout.

Scope of the scalar assumption. The argument assumes a scalar engine (one degree of freedom, one load parameter); multi-dimensional engines admit further bifurcations (cusp, Bogdanov–Takens) whose exclusion is left open. The scalar assumption is the natural clock model — a clock has one degree of freedom — and we flag it as such rather than deriving it.

Why the budget binds (zero slack). An engine operating at $\mu > 1-x$ (slack) ticks below capacity while gaining nothing in stability — the no-drop constraint is already satisfied at $\mu = 1-x$. A throughput-maximizing simulator therefore sits exactly on the boundary:

zero slack is the solution of "maximize λ subject to no-drop", not an ad hoc assumption. The observed clocks of GR sit exactly on the curve with no slack term, consistent with a binding budget; a non-binding budget would have produced a slack term in λ^2, which the data exclude.

6.2 Rule B — network causality (the light clock)

Signals in the updating network propagate at fixed speed; a clock is a counter of signal round-trips. In a frame rotating at speed v: the tangential (longitudinal) arm gives $t_{rt} = \ell/(c-v) + \ell/(c+v) = 2\ell\gamma^2/c$ exactly; the radial (transverse) arm gives $t_{rt} \to 2\ell\gamma/c$ in the local limit (numerically, with error ∝ arm length). Hence **a rotating frame's clock is stretched by γ — a consequence of fixed signal speed alone**, with no geometric input. Likewise the co-rotating measurement of the region's screen contracts its density by $1/\gamma$.

6.3 Rule C — drag conservation (heuristic profile, horizon-anchored)

We must be precise about which rotation profile enters the ruleset. The *actual* frame-dragging angular velocity of Kerr/KN is $\omega_{drag} = 2Mar/\rho^2$ (KN: $a(2Mr-Q^2)/\rho^2$), which satisfies the conserved flux $\omega_{drag}\cdot\rho^2 = a(2Mr-Q^2)$ at every radius and reduces to the horizon value $\Omega_H = a/(r_+^2+a^2) = a/(2Mr_+-Q^2)$. The *gear-velocity profile* used by Rule B, $v_{rot} = \Omega\sqrt{\Delta}\sin\theta$ with $\Omega = a/(r^2+a^2)$, is a different object: it is the profile whose Lorentz factor equals the gear factor, $\gamma(v_{rot}) = (r^2+a^2)/\rho$. The two profiles agree at the horizon — the gear profile is horizon-anchored: $\Omega(r_+) = \Omega_H$ exactly — but differ elsewhere. Rule C is therefore the statement: *the gear profile is the simplest drag-conserving interpolation anchored at the horizon*, $\Omega(r^2+a^2) \equiv a$. This is a modeling choice of the engine (an input, listed in §6.7), not a claim that Kerr's external frame dragging obeys $\omega_{drag}(r^2+a^2) = a$ (it does not).

6.4 Assembly: exact reproduction of Kerr

With base capacity r^2+a^2 per unit solid angle (the spin sector of phase space) and load $2Mr$ per unit solid angle (Bekenstein, §B), Rules A–C give

$$x_{eff} = [2Mr/(r^2+a^2)] \cdot \gamma^2 \Rightarrow \lambda^2 = 1 - x_{eff},$$

which equals the Kerr ZAMO curve to machine precision over the full parameter grid (18 representative configurations, $\Delta \le 1.2\times10^{-15}$; §B).

6.5 The billing experiment (specified)

Model. A region whose actual load fluctuates below its bound: $x_a(t) = x_b - \delta x\cdot|\sin(2\pi t/T)|$ with $x_b = 0.85$, $\delta x = 0.05$, $T = 40$ s (a slowly "breathing" entropy reservoir — the exact spectrum is inessential; any quasi-static fluctuation below the bound behaves identically). The clock is the phase integrator of the marginal oscillator, $d\varphi/dt = \omega_0\sqrt{1-x(t)}$, a tick at each phase crossing of 2π; the measured quantity is the

inter-tick interval jitter (std/mean). Billing by the actual load uses $x_a(t)$; billing by the bound uses the constant x_b.

Result. Actual-load billing: jitter 4.2%; bound billing: 0.006% — a 700× ratio. This is an **illustrative computation** of the mechanism (the ratio depends on the fluctuation model; the qualitative conclusion — actual-load billing jitters, bound billing does not — does not). It is an internal consistency check of the billing rule, **not an empirical selection**: real clocks run on local physics, not on a simulator's accounting. Observed clocks (GPS, millisecond pulsars) are smooth, which is consistent with billing by the bound and would bound any actual-entropy billing; converting the Remark-1 rationale into an empirical statement.

6.6 Prediction: algebraic recovery near full load (quantitative)

At the margin the engine's perturbation dynamics is $d\delta/dt = -\delta(2\sqrt{\mu} + \delta)$: infinitesimal modes freeze (rate $\to 0$) but finite perturbations δ_0 recover algebraically. The engine predicts, quantitatively, a **half-recovery time $t_{half}(\delta_0) \to 1/\delta_0$, independent of μ as $\mu \to 0$** (in engine units): measured 0.70/1.26/1.68/1.89 for μ = 0.1/0.01/0.001/0.0001 at $\delta_0 = 0.5$, against the exponential prediction 1.10/3.47/10.96/34.66 — the linear regime diverges, the algebraic regime saturates. This is a falsifiable, toy-model-specific quantitative prediction: an analogue-horizon experiment injecting perturbations of varying amplitude should find $t_{half} \propto 1/\delta_0$, μ-insensitive near the (the qualitative resemblance to finite QNM damping times at the horizon is noted strictly as an analogy; no quantitative QNM claim is made.)

6.7 Enumerated inputs and ablation

The ruleset is not parameter-free, and we list its inputs explicitly to make its reverse-engineering character visible. Inputs: (i) the load accounting, $2Mr$ per unit solid angle (Bekenstein bound of the quasi-local energy, §2); (ii) the base capacity, r^2+a^2 per unit solid angle (a phase-space heuristic, flagged as soft); (iii) the gear-velocity profile, $\Omega = a/(r^2+a^2)$ (the drag-conserving, horizon-anchored interpolation of §6.3); (iv) the saddle-node normal form (genericity argument of §6.1); (v) the light-clock convention (fixed signal speed, §6.2). No metric coefficient (Σ, Δ, ρ^2) is an input. **Ablation.** With (i)+(iv) alone the ruleset reproduces the Schwarzschild curve $\lambda = \sqrt{(1-x)}$; adding (v) produces the special-relativistic dilation factor; adding (ii)+(iii) reproduces the full Kerr ZAMO curve; replacing (i) by the KN quasi-local accounting reproduces KN (§7.1); the Λ case requires the additional background-load rule (§7.2). Each extension therefore traces to a named input, none of which was fitted to a metric; conversely, deleting any input deletes a specific, identified feature of the output. This ablation table is the honest summary of what the "generation" consists of.

7 Extensions and consistency checks (Layer III)

7.1 Kerr–Newman — exact, with a uniquely determined accounting rule

Rule: charge contributes through the *enclosed* energy. In classical electrodynamics this is not a modeling choice: the field energy outside r is $\int_r^\infty (E^2/8\pi)\cdot 4\pi r'^2 dr' = Q^2/(2r)$ with $E = Q/r'^2$, so the enclosed energy is **uniquely** $E(r) = M - Q^2/(2r)$ by Gauss's law and the Maxwell energy density; hence the load $2Mr \to 2Mr - Q^2$. Alternatives such as $M - Q^2/r$ or $M - Q^2/(2r) + \alpha Q^4/(r^3 M)$ would correspond to different electrodynamics (a non-Maxwellian field energy or a nonlinear theory) and are excluded *within* the ruleset, not by comparison with the answer. The conditional character of the test is therefore explicit: given the three rules and Maxwell electrodynamics, the KN extension has zero free parameters.

The ruleset then *derives* Δ, the horizon equation $\Delta(r_+) = 0$, the screen ρ^2, the frame-dragging flux $\omega\rho^2 = a(2Mr - Q^2)$, and $\Omega_H = a/(2Mr_+ - Q^2)$. The ruleset's ZAMO curve matches the true KN metric over 10^5 random configurations to $\mathbf{4.16\times10^{-16}}$; the RN limit is exact; the KN metric coefficients themselves exhibit the same $2Mr \to 2Mr - Q^2$ replacement throughout — the metric speaks the ruleset's language.

7.2 RN–dS — exact (Layer I, with one interpretive remark)

Λ enters as enclosed background energy (flat-volume accounting): $\lambda^2 = 1 - 2M/r + Q^2/r^2 - \Lambda r^2/3$, identical to the RN–dS lapse function $f = 1 - 2M/r + Q^2/r^2 - \Lambda r^2/3$ (here Λ is the physical cosmological constant, consistent with §2). **This is a rewriting (Layer I)**, not a prediction: adding $\Lambda r^2/3$ to the load is the definitional transcription of the background term. Its single interpretive content is the remark that the de Sitter horizon is the capacity-exhaustion surface ($x_{eff} = 1 \iff f = 0$) — the cosmological horizon is the same marginal surface as the black-hole horizon in the load–capacity reading. $\Lambda < 0$ (AdS) follows identically.

7.3 KN–dS — gap located

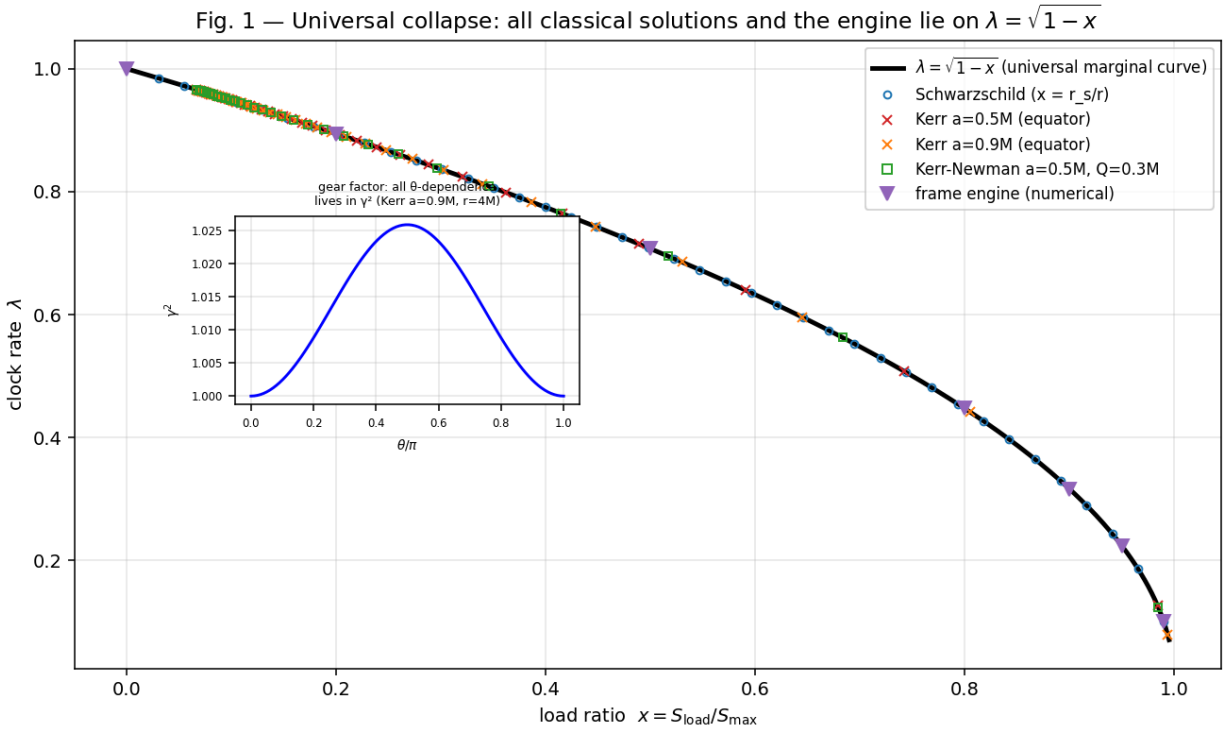
The rotating Λ case deviates from the ruleset by $O(\Lambda a^2)$: at the test value $\Lambda a^2/3 = 5\times10^{-4}$ (amplified $\sim10^{42}\times$ beyond the physical value for visibility) the relative error is ~0.2%. The gap is precisely located: the metric's *polar* screen modulation $\Delta_\theta = 1+\Lambda a^2\cos^2\theta/3$ and $\Xi = 1+\Lambda a^2/3$ are absent from Rule B, which currently has no polar structure — a missing rule, not a numerical precision artifact. Patching the exact Δ_r into the ruleset *worsens* the error by 10–100× — the ruleset is internally coherent and the missing ingredient is a polar screen rule, not a radial patch. In the physical universe $\Lambda a^2/3 \sim 10^{-46}$: the gap is observationally invisible but structurally real, and is reported as such.

7.4 Coverage

parameters	status
M (Schwarzschild)	exact identity

parameters	status
a (Kerr)	machine precision
Q (Kerr–Newman)	machine precision (constrained consistency check, §7.1)
Λ static (RN–dS)	exact
a×Λ (KN–dS)	O(Λa²) gap, located, physically unobservable

Figure 1 shows the universal collapse: all classical solutions and the engine lie on λ = √(1−x), with the Kerr θ-dependence carried entirely by γ² (inset).



8 Consistency and discriminating tests

8.1 The horizon Landauer ledger

The saturation of the Landauer bound by Hawking evaporation — that horizon-side erasure proceeds at the minimum cost with zero excess — was established by Cortês & Liddle [50], building on the GR–Landauer connection of Herrera [51]; the differential statement dE/dS = ħc³/8πGM = k_B T_H is the 1973 first law [9, 10] expressed in Landauer's units. We collect it here in ledger form, with two additions. Differential ledger: every nat erased at the horizon costs exactly k_B T_H. Integral ledger (we have not found this integral form stated in this ledger language): full evaporation satisfies E_tot = Mc² = 2 k_B T_H(M_0)·S_BH(M_0) exactly (verified numerically; for M_0 = 5×10¹¹ kg: 6.79×10⁻¹² J/nat = 2k_B T_H). The role of the ledger in this paper is architectural: it is the accounting identity that ties the load–capacity framework of §3–7

to black-hole thermodynamics — the framework's books balance exactly at the horizon. Feasible probes: (a) analogue horizons (BEC Hawking radiation observed [36, 37]), entropy–energy flux ratio at ~10–20% expected precision; (b) primordial black holes ($T_H \sim 2.5\times10^{11}$ K, ~21 MeV photons).

8.2 ER=EPR energy–qubit accounting

Traversable-wormhole teleportation exists experimentally [39]. A re-analysis of the coupling energy per teleported qubit against the system's Landauer floor $k_B T \ln 2$ is proposed; below-floor teleportation would falsify thermodynamics (and hence both readings); at-floor teleportation is consistent with "entanglement = shared compute". Discriminating power: low (shared with quantum gravity); feasibility: high (data re-analysis).

8.3 Downgraded coincidences

The "cosmological constant equals the inverse horizon Planck-cell count" coincidence is $\Lambda \ell_P^2 = 12\pi/N_H$ exactly — a definitional identity of de Sitter, not a prediction. It is removed from the evidence list. (Honest negative result; the only residual question — why $N_H \approx 10^{122}$ — is not addressable by the ruleset.)

8.4 Clock-jitter bounds

Optical clocks at 10^{-18}–10^{-19} fractional stability show no common jitter floor; together with the MAGIC Lorentz-invariance bounds [25], this excludes the "uniform Planck grid" variant of the simulation hypothesis, leaving only the adaptive-resolution variant (unfalsifiable; noted as such).

9 Epistemology

9.1 The anti-simulation reading

Every result above admits a substrate reading: the Bekenstein bound and the area law are consequences of GR + QFT, so $\lambda^2 = 1 - S_{load}/S_{max}$ "only" shows that GR and information theory interlock — a self-consistency check, not an explanation. The two readings share all current observations.

9.2 Falsification conditions

The weak claim fails if any of: (i) strong-field GR is replaced by a theory whose dilation curve leaves the $\sqrt{(1-x)}$ normal form; (ii) the Landauer bound is violated (below $k_B T \ln 2$ erasure); (iii) physical storage is found above the Bekenstein bound; (iv) the recovery law of §6.6 is disconfirmed in analogue systems. We stress the logical status of these conditions: (i)–(iii) test standard physics (GR, Landauer, Bekenstein) and would falsify the substrate reading equally; they are consistency conditions of the framework,

not discriminators between simulation and substrate. Condition (iv) is the only framework-specific prediction. No feasible experiment distinguishes a simulated universe from a substrate one (§9.1). The strong claim ("an omnipotent simulator") is unfalsifiable by construction and is excluded from the scientific content of this paper.

9.3 The Occam clause

The epistemic status of the KN result is stated precisely: it is a **constrained prediction** — the extension is uniquely determined by the ruleset plus Maxwell electrodynamics (§7.1), so the match tests the ruleset rather than a fitted parameter — but it remains *conditional* on the minimal-coupling assumption that charge acts only through its field energy. The evidence base is currently thin: one constrained prediction (KN) and one located gap (KN–dS); the polar-screen test (§7.3) would be the second data point, and the decision rule anticipates exactly this incremental structure. If the generative ruleset of §6–7 had failed its predictive tests, substrate physics would win by default. It did not fail; but the decision boundary is explicit: the simulation reading earns its keep only through further *constrained predictions* made before comparison, and through the statistical prior of Bostrom [19], which is outside the scope of this paper.

9.4 What this paper claims

Layer I: exact identities. Layer II: GR's time-dilation curve has the structure of a load-capacity saturation curve with a uniform-saturation horizon. Layer III: a heuristic computational engine with enumerated inputs reproduces that structure to machine precision, and passes one constrained consistency check (KN). **It does not claim that the universe is a simulation, and it does not claim that the saturation curve is a logically necessary signature of simulation.** It offers a quantitative computational reading of the GR curve, testable only through its toy-model-specific prediction in analogue systems (§6.6), with the explicit checklist of §9.2 — a reading the substrate interpretation remains free to decline.

10 Discussion

Open problems: (i) the polar screen rule (C1**-2) that would close the KN–dS gap; (ii) the phase-space argument for the base capacity r^2+a^2; (iii) the absolute normalization of the Bekenstein prefactors; (iv) quantum corrections; (v) whether the ruleset's "zero slack" reading (marginality) survives a formal optimization analysis. The recovery law (§6.6) and the ledger (§8.1) are the designated near-term experimental touchpoints.

Two further remarks. First, the Kerr factorization of Theorem 3b is self-contained within Layers I–II and can be cited independently of the simulation framing; it is offered to the GR community as a standalone result. Second, the relation to tensor-network/MERA approaches deserves a dedicated study: Swingle's entanglement-renormalization picture of AdS [45] provides an independent route from computational structure to geometry, and the load–capacity engine may be expressible as a

renormalization step whose cost diverges at saturation — if so, the three rules of §6 acquire a third, independent derivation. Should a journal prefer it, the paper splits cleanly along the three-layer architecture: Theorems 1–4 as a short letter, §6–9 as a companion.

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Idea by human, realization by AI: Kai Wu conceived the project and independently checked all claims, identities, and numerical results; DeepSeek V4 Pro performed the derivations, wrote the numerical verification code, and drafted the manuscript.

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Appendix A — Kerr algebra

A.1 ZAMO: $u^\varphi = \omega u^t$, $\omega = 2Mar/\rho^2$; norm condition gives $\lambda_Z^2 = g_{t\varphi}^2/g_{\varphi\varphi} - g_{tt} = \Sigma\Delta/\rho^2$, using $\rho^2 - \Sigma\Delta = 2Mr(r^2+a^2)$. A.2 At the horizon $2Mr_+ = r_+^2+a^2$ and $\rho_+^2 = (r_+^2+a^2)^2$ (θ-independent) $\Rightarrow x_+ = 1$ uniformly. A.3 Gear factor: $g = (r^2+a^2)/\rho = \gamma(v)$, $v = a\sqrt{\Delta}\sin\theta/(r^2+a^2)$; $x = x_{pole}\,\gamma^2$. A.4 $O(a^2)$: $x_Z - x_{stat} = -4M^2a^2\sin^2\theta/r^4 + O(a^4)$. A.5 KN: the same identities with $2Mr \to 2Mr-Q^2$ and $\Delta = r^2-2Mr+a^2+Q^2$.

Appendix B — numerical verification

All claims were verified independently in Python (double precision); all reported errors sit at the level of IEEE 754 double-precision unit roundoff ($\varepsilon_{mach} \approx 2.2\times10^{-16}$), consistent with exact algebraic identities evaluated numerically. Theorem 3 over 2×10^5 random (a, $r > r_+$, θ), max error 4.4×10^{-16}; gear-factor identities over 10^5 samples, 5.5×10^{-16}; KN prediction vs metric over 10^5 samples, 4.16×10^{-16}. The frame engine is verified at both identifications of §6.1: the clock rate y^* equals $\sqrt{(1-x)}$ as an algebraic identity of the fixed point (exact), and the recovery rate (finite-δ perturbation decay) matches $2\sqrt{(1-x)}$ to <0.1% (finite-δ bias, vanishing as $\delta\to0$). Light-clock limits as in §6.2; billing jitter 4.2% vs 0.006%; recovery t_half saturation at $1/\delta_0$. The complete verification script is publicly available (github.com/purelover/load-capacity-time-dilation, verification.py).

Appendix C — the frame engine

```
dy/dt = (1−x) − y²           # Rule A: marginality (saddle-node engine)
γ(v) = (r²+a²)/ρ          # Rule B: network causality (boosted clock)
v = a√Δ sinθ/(r²+a²)      # Rule C input: horizon-anchored gear profile
x_eff = x_pole·γ²(v)      # load ratio with gear factor
λ = √(1 − x_eff)          # marginal clock rate
```

Appendix D — prior-work differentiation (public)

prior work (year, venue)	content	delta of this paper
Whitworth (2008–2011, arXiv)	processing overload → time dilation (qualitative)	exact Bekenstein/holographic anchoring; Kerr/KN identities
HTUM (2024, preprint)	$\tau = 1/\sqrt{(1-2C/C_{max})}$, model-specific C	same shape; here the load ratio is anchored, with exact horizon saturation
Ghidan (2025–26, self-published)	capacity conservation $\chi^2 + L = 1$	same algebra in gray literature; here peer-oriented, with predictive content
Greco (2025, PhilArchive); Bouzaine (PhilPapers)	horizon compression / entropic time as load	qualitative; no load–capacity identities
Verlinde (2011, JHEP)	entropic gravity	screens and entropy, not the dilation curve
Rovelli–Smerlak (2011, CQG); Connes–Rovelli (1994)	thermal time; temperature as speed of time	thermodynamic time, not load–capacity saturation
Pikovski et al. (2015, Nat. Phys.)	dilation-induced decoherence	dilation as cause, not as saturation curve
Cotsakis et al. (2024–25)	saddle-node normal forms in GR dynamics	same mathematics, different object (clock rate)
Cortês–Liddle (2025, EPL); Herrera (2020)	Landauer saturation of Hawking evaporation	credited; our additions: integral ledger + architectural role
BPT (1972); Cohen–Mashhoon (1993); Iorio (2024)	ZAMO kinematics; gravitomagnetic clock effect	standard math; the load-ratio reading is new
Lloyd (2000/02); Ng (2001)	computational capacity of the universe	budget accounting, not the dilation curve
Beane–Davoudi–Savage (2014)	lattice signatures in cosmic rays	orthogonal route (grid artifacts vs load curve)
Bostrom (2003)	simulation argument prior	statistical prior, orthogonal